Spot Detection from Averaged TIFF Frames

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Objective and workflow

Goal

Detect point-like emitters in averaged TIFF image stacks and determine whether each bright connected region contains **one spot** or **two overlapping spots**. The analysis is applied separately to **50 ms** and **100 ms** exposure datasets.

Processing pipeline

1. Load cropped TIFF frames and compute the averaged image for each exposure setting.
2. Apply an intensity threshold to generate a binary mask.
3. Label connected components and extract local ROIs around each candidate region.
4. Fit both **1-Gaussian** and **2-Gaussian** models to each ROI.
5. Use **BIC** and minimum spot separation to decide whether a region contains one emitter or two.

Preprocessing: Stack Averaging

Stack averaging

$$\bar{I}(x, y) = \frac{1}{N} \sum_{k=1}^{N} I_k(x, y)$$

- Frames are summed and averaged

Notes:

- Exposure time (50 ms vs 100 ms)
- Stacked 100 images for 100ms
- Stacked 200 images for 50ms

Thresholding and Connected Components

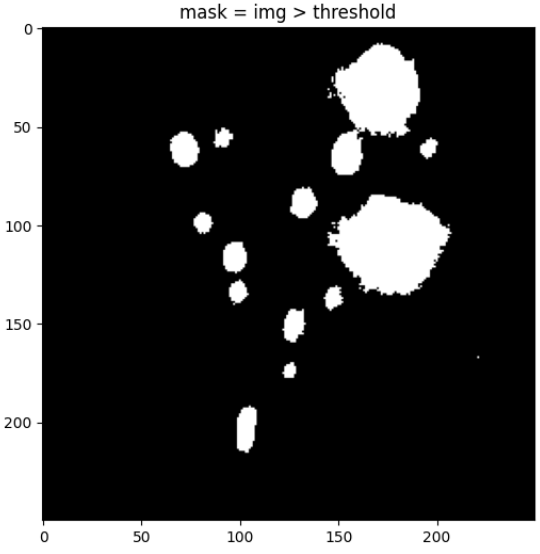
Binary mask

$$M(x, y) = \mathbf{1}[I(x, y) > 8]$$

- Fixed threshold $= \mathbf{8}$
- Operates on averaged image

Component detection

- Connected regions via `ndi.label`
- Area filter:
  - Min $= 3$ pixels
  - Max $= 10{,}000$ pixels
- ROI expanded by **4 pixels**



Gaussian Fitting and Model Selection

Models per component

Single Gaussian:

$$G_1 = A_1 e^{-\frac{(x-x_1)^2+(y-y_1)^2}{2\sigma_1^2}} + B$$

Two Gaussians:

$$G_2 = G_A + G_B + B$$

Fitting procedure

- Nonlinear least squares (`curve_fit`)
- Fit both 1-spot and 2-spot models
- Compute residual sum of squares (RSS)

BIC selection

$$\mathrm{BIC} = n \ln\left(\frac{\mathrm{RSS}}{n}\right) + k \ln(n)$$

Choose 2-spot model only if:

$$\mathrm{BIC}_2 < \mathrm{BIC}_1 - 25$$

and separation $\geq$ **5 pixels**

Why this works

The two-spot model is only accepted if it significantly improves the fit while avoiding overfitting, and if the detected spots are sufficiently separated

Model Selection via RSS and BIC

Residual error (fit quality) between the pixel intensities in the ROI vs the Gaussian model's predicted intensities

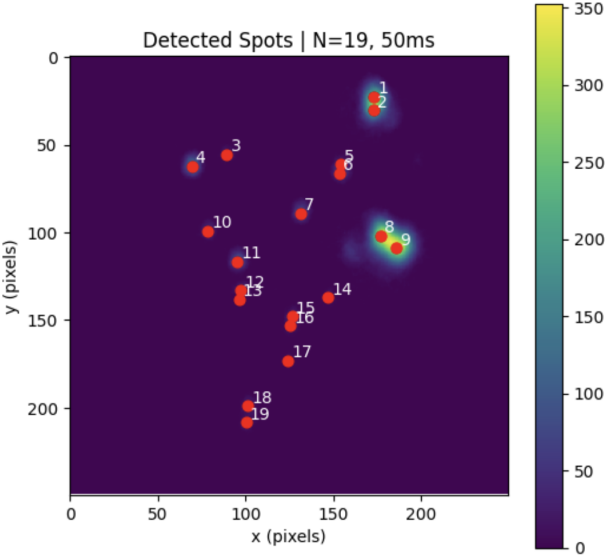
$$\mathrm{RSS} = \sum (\text{data} - \text{model})^2$$

Bayesian Information Criterion (BIC)

$$\mathrm{BIC} = n \ln\left(\frac{\mathrm{RSS}}{n}\right) + k \ln(n)$$

- First term: goodness of fit
- Second term: penalty for model complexity

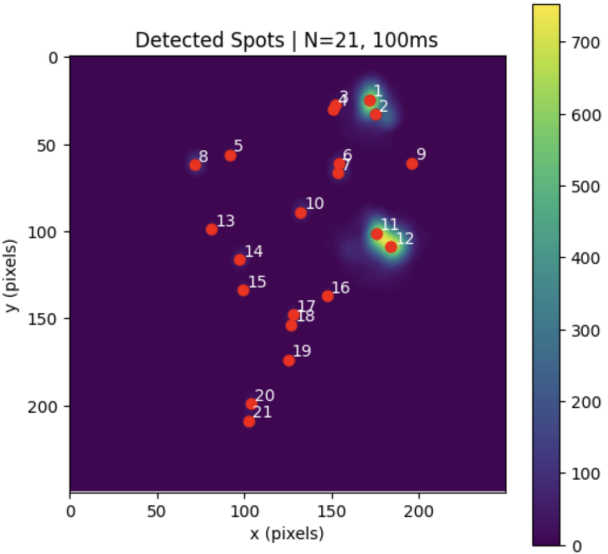
Detection results and next steps 1



Key observations

- 50ms image
- Final count: **19 centroids**
- Larger bright regions can split into multiple emitters when the 2-Gaussian model is fit.
- The code checks close-pair distances for **(11,13)** and **(15,16) and (18,19)**.

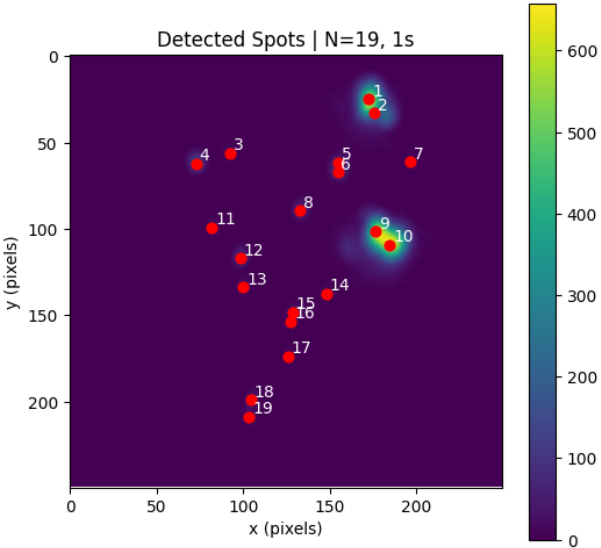
Detection results and next steps 2



Key observations

- 100ms image
- Final count: **21 centroids**
- Larger bright regions can split into multiple emitters when the 2-Gaussian model is justified.
- The code checks close-pair distances for **(14,15)** and **(17,18) and (20,21)**.

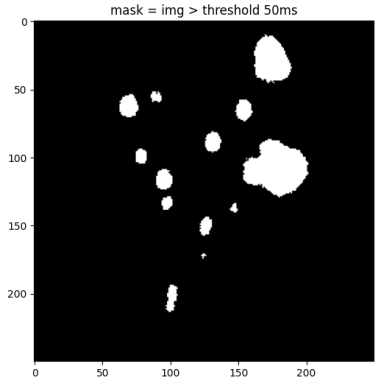
Detection results and next steps 3



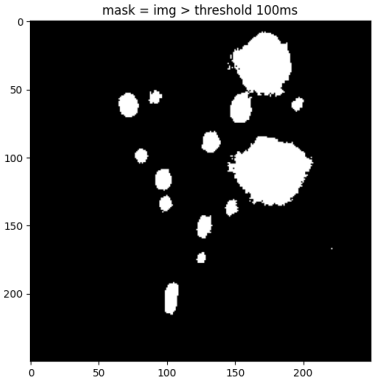
Key observations

- 1s image
- Final count: **19 centroids**
- Larger bright regions can split into multiple emitters when the 2-Gaussian model is justified.
- The code checks close-pair distances for **(12,13)** and **(15,16) and (18,19)**.

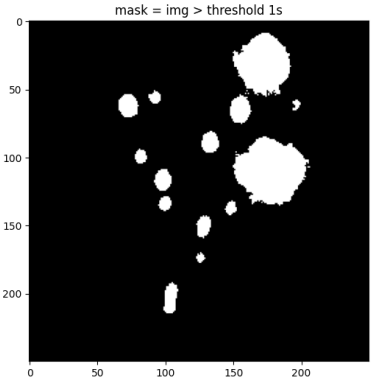
Thresholding Comparison Across Exposure Times



50 ms



100 ms



1 s

Binary masks obtained via thresholding at different exposure times. Increasing exposure leads to larger and more saturated spot regions, affecting component segmentation and downstream centroid fitting.